

little more than a screen connected to the Internet. Almost all of their value is determined by their software. Even old-school products such as automobiles are seeing ever-larger parts of their value being generated by the software they carry inside, which controls everything from the entertainment system to tuning the engine to controlling the brakes. What can be built out of software can be modified much faster than a physical or mechanical device can.

2. Fast production changes. Because of the success of the lean manufacturing movement, many assembly lines are set up to allow each new product that comes off the line to be customized completely without sacrificing quality or cost-effectiveness. Historically, this has been used to offer the customer many choices of product, but in the future, this capability will allow the designers of products to get much faster feedback about new versions. When the design changes, there is no excess inventory of the old version to slow things down. Since machines are designed for rapid changeovers, as soon as the new design is ready, new versions can be produced quickly.

3. 3D printing and rapid prototyping tools. As just one example, most products and parts that are made out of plastic today are mass produced using a technique called injection molding. This process is extremely expensive and time-consuming to set up, but once it is up and running, it can reproduce hundreds of thousands of identical individual items at an extremely low cost. It is a classic large-batch production process. This has put entrepreneurs who want to develop a new physical product at a disadvantage, since in general only large companies can afford these large production runs for a new product. However, new technologies are allowing entrepreneurs to build small batches of products that are of the same quality as products made with injection molding, but at much lower cost and much, much faster.